

3(a)	(insect) vector / <i>Anopheles</i> / (a) mosquito ; <i>Plasmodium, falciparum / ovale / malariae / vivax</i> ; A <i>knowlesi</i> <i>Vibrio cholerae</i> and cholera ; A other pathogens and associated diseases that are transmitted in the same way <i>Mycobacterium, tuberculosis / bovis</i> ; airborne droplets / droplet infection / aerosol (infection) ; I air droplets I water droplets from, coughing / sneezing A alternative transmission only if <i>Mycobacterium bovis</i> stated e.g. (eating) contaminated meat (from infected cattle) (drinking) contaminated milk (drinking) unpasteurised milk from contaminated, cows / cattle	5
------	---	----------

Question	Answer	Marks
3(b)	any two from: tenofovir competes with, (activated / phosphorylated) adenine nucleotide / deoxyribose adenosine triphosphate / dATP, for active site / to prevent nucleotides being added to elongating chain ; A (tenofovir acts as a) competitive inhibitor (of reverse transcriptase) tenofovir forms a phosphodiester bond to elongating DNA strand, but stops further reactions / AW ; tenofovir has, no 3' -OH so next nucleotide cannot form a phosphodiester bond / no (deoxy)ribose so cannot form phosphodiester bond ; A pentose R sugar unqualified	2
3(c)(i)	20 (%) ; R answers with decimal places	1

Question	Answer	Marks
3(c)(ii)	<i>there must be at least one statement or at least one explanation to gain max 4, otherwise mark to max 3</i> any four from: 1 supply, condoms / femidoms / dental dams / item(s) for protection during sex ; 2 barrier to transmission during sexual intercourse ; 3 <i>ref. to</i>, needle exchange schemes / other suitable support for intravenous drug (ab)users ; 1 clean needles 4 decreases risk of sharing contaminated equipment / AW ; 5 use, new needles / new syringes / sterilised equipment / AW, for medical procedures ; 1 clean needles 6 decreases risk of transmission from contaminated blood ; 7 provide testing for HIV, in high risk groups / to individuals at high risk ; 8 for early diagnosis so newly infected people so can start drug treatment immediately ; 8 test pregnant women for HIV / provide powdered milk to women who are HIV positive ; 9 prevent women who are HIV positive passing HIV in breast milk ; 10 prevent people who are HIV positive being blood donors / screen donated blood / heat-treat donated blood ; 11 prevent people receiving blood infected with HIV, during blood transfusions / operations ; 12 carry out contact tracing ; 13 locate people who, are undiagnosed / may be HIV positive / should be offered test ; 14 supply (named) drug(s) to people, living with HIV / who are HIV positive / pregnant women with HIV ; 15 prevents HIV, spreading throughout the body / infecting more T-lymphocytes ; A to reduce viral load 16 provide, education / information, about, HIV treatments / HIV transmission ; 17 to raise awareness of ways to reduce infection / AW ; A use of barrier methods during sex / safer sex 18 AVP ; e.g. law to make it illegal to knowingly transmit the virus	4